



Voltage Equations

d-q axis voltage equation: GOAL

- GOAL: 아래 수식을 직접 유도할 수 있을 것.

$$\begin{cases} V_{ds}^r = R_s i_{ds}^r + L_s \frac{di_{ds}^r}{dt} - \omega_r L_s i_{qs}^r \\ V_{qs}^r = R_s i_{qs}^r + L_s \frac{di_{qs}^r}{dt} + \omega_r L_s i_{ds}^r + \omega_r \phi_f \end{cases}$$

- Definitions:

$L_{\{s\}}$: phase inductance

$R_{\{s\}}$: phase resistance

$\omega_{\{r\}}$ or $\omega_{\{e\}}$: Electrical velocity of rotor frame

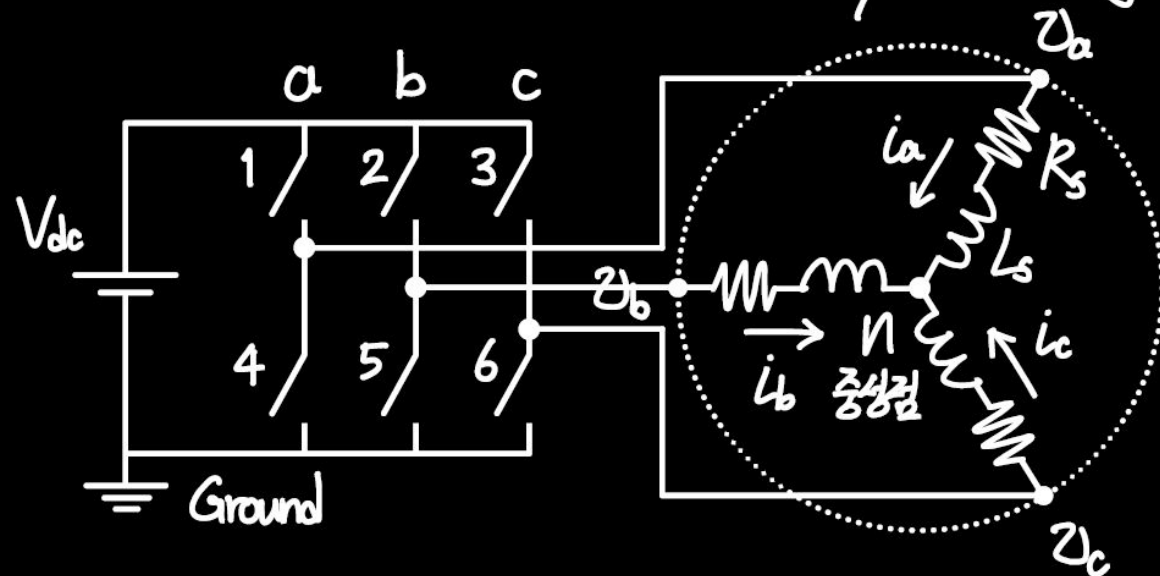
- [Quiz]

오직 오실로스코프를 통해 주어진 SPMSM의

토크 상수를 측정하는 방법과 그 공식을 제안하라.

d-q axis voltage equation: Modeling

□ Modeling of PMSM



Star(Wye) Winding. Low current, High Voltage

v_a, v_b, v_c : 극전압 (Terminal voltage)

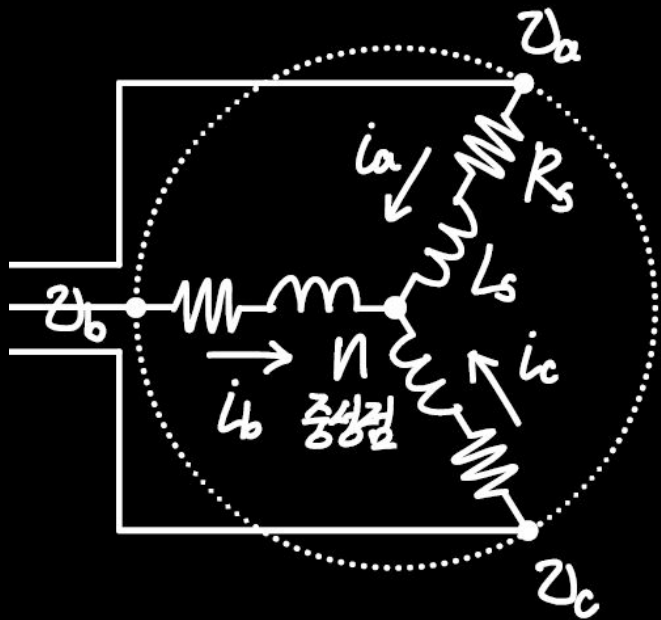
i_a, i_b, i_c : 상전류 (Phase current)

v_{as}, v_{bs}, v_{cs} : 상전압 (Phase voltage)

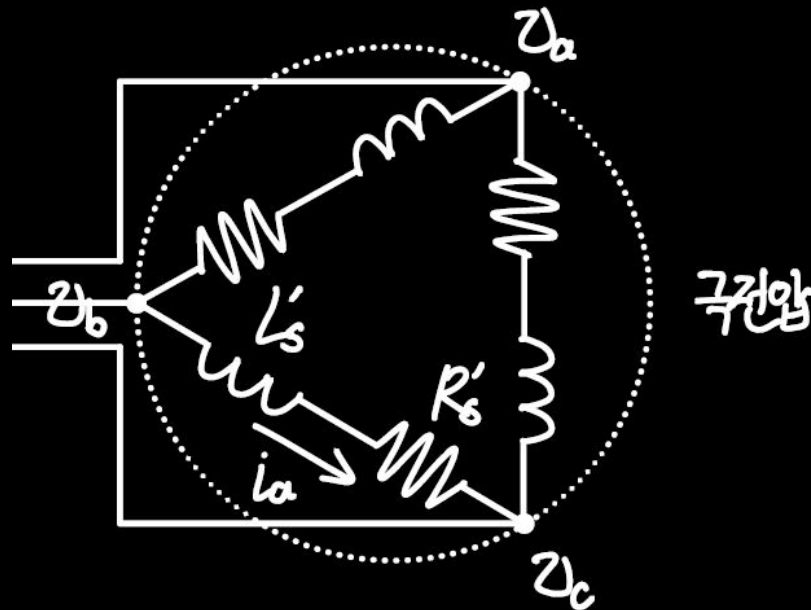
$v_{as} \triangleq v_a - v_n$ ✖ v_n 은 평균적으로 ground 전위에 해당할 수 있으나, ground와 전기적으로 무리함 도선으로 연결되어있지 않다.

- 정의에 매우 유의할 것!!!

d-q axis voltage equation: Wye/Delta Winding Equivalent Circuit



- Wye/Delta는 등가 회로로 나타낼 수 있다.



$$\begin{aligned} (\text{star}) 2L_s &= \frac{2}{3} L'_s (\text{delta}) = 2L_T (\text{delta, Terminal}) \\ (\text{star}) 2R_s &= \frac{2}{3} R'_s (\text{delta}) = 2R_T (\text{delta, Terminal}) \end{aligned}$$

- [Quiz]

ABC 3상 단자를 통해 해당 stator가 Delta/Wye Winding인지 구분할 수 있는가?

d-q axis voltage equation: ABC Phase Voltage Equation

$$v_{as} = R_s i_{as} + \frac{d\lambda_{as}}{dt}$$

rotor magnet에 의한 flux

$$\lambda_{as} = \lambda_{aa} + \lambda_{ab} + \lambda_{ac} + \phi_{af} = L_{aa} i_a + L_{ab} i_b + L_{ac} i_c + \phi_{af}$$

$$v_{bs} = R_s i_{bs} + \frac{d\lambda_{bs}}{dt}$$

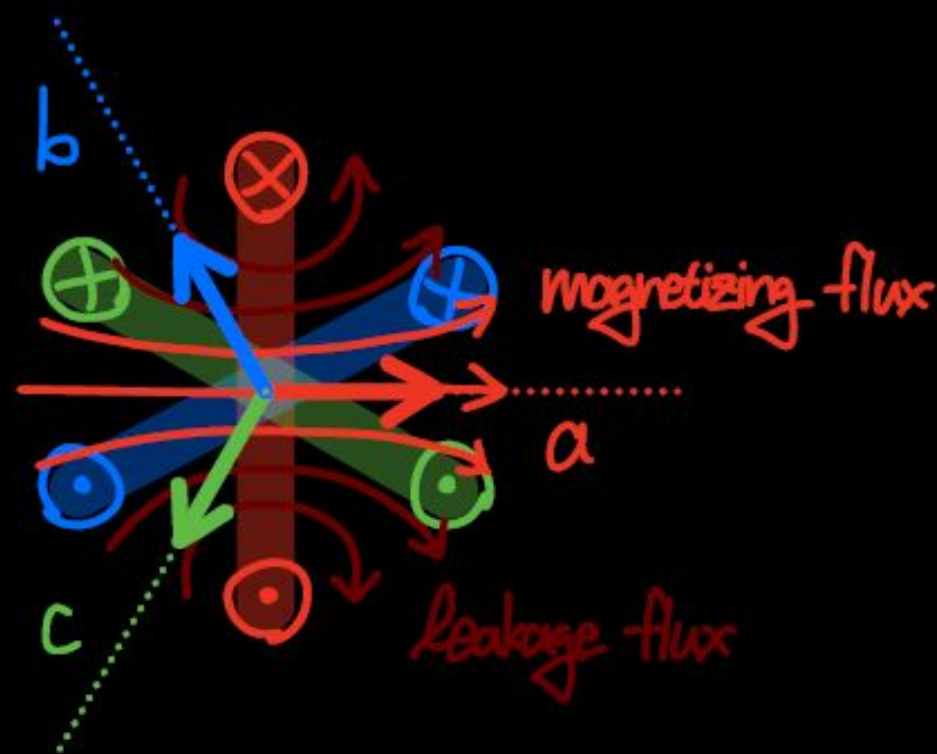
$$\lambda_{bs} = \lambda_{ba} + \lambda_{bb} + \lambda_{bc} + \phi_{bf}$$

$\lambda_{af} \triangleq$ 권선 y에 의해 발생하여 권선 z에 쇄교하는 자속

$$v_{cs} = R_s i_{cs} + \frac{d\lambda_{cs}}{dt}$$

$$\lambda_{cs} = \lambda_{ca} + \lambda_{cb} + \lambda_{cc} + \phi_{cf}$$

$$v_{abcs} = R_{3 \times 3} i_{abcs} + \frac{d\lambda_{abcs}}{dt}$$



d-q axis voltage equation: Flux Equation

• [Quiz]

왜 삼각함수인가?

L_{ls} : leakage ind. (아상각자, 0) L_{ms} : magnetizing ind.

$$\begin{bmatrix} \lambda_{as} \\ \lambda_{bs} \\ \lambda_{cs} \end{bmatrix} = \begin{bmatrix} L_{ls} + L_{ms} & -\frac{1}{2}L_{ms} & -\frac{1}{2}L_{ms} \\ -\frac{1}{2}L_{ms} & L_{ls} + L_{ms} & -\frac{1}{2}L_{ms} \\ -\frac{1}{2}L_{ms} & -\frac{1}{2}L_{ms} & L_{ls} + L_{ms} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} \phi_f \cos \theta_e \\ \phi_f \cos(\theta_e - \frac{2\pi}{3}) \\ \phi_f \cos(\theta_e + \frac{2\pi}{3}) \end{bmatrix}$$

• [Quiz]

왜 $-(1/2) * L_{ms}$ 인가? 항상 그럴까?

• [Quiz]

Inductance L_{xs} 는 rotor의 각변위 θ_e 에 대해 constant인가?

d-q axis voltage equation: Park-Clark Transformation

$$T(\theta) = \frac{2}{3} \begin{bmatrix} \cos\theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{4\pi}{3}) \\ -\sin\theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta - \frac{4\pi}{3}) \end{bmatrix}$$

↑ Magnitude invariance.

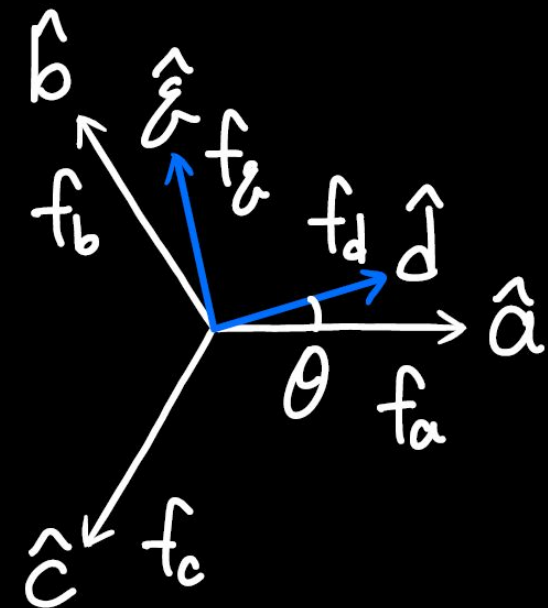
$$T_{inv}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \cos(\theta - \frac{2\pi}{3}) & -\sin(\theta - \frac{2\pi}{3}) \\ \cos(\theta - \frac{4\pi}{3}) & -\sin(\theta - \frac{4\pi}{3}) \end{bmatrix}$$
$$T(\theta)T_{inv}(\theta) = I_{2 \times 2}$$

$$\mathcal{V}_{dqs}^\theta = T(\theta) \mathcal{V}_{abcs}$$

$$\mathcal{I}_{dqs}^\theta = T(\theta) \mathcal{I}_{abcs}$$

- [Quiz]

d-q axis 변수들은 다른 convention에 의해 다르게 정의 가능한가?



d-q axis voltage equation: Voltage Equation – rotor frame

rotor 좌표계 θ_e 에 대해,

$$\begin{cases} V_{ds}^r = R_s i_{ds}^r + \frac{d\lambda_{ds}^r}{dt} - \omega_r \lambda_{qs}^r \\ V_{qs}^r = R_s i_{qs}^r + \frac{d\lambda_{qs}^r}{dt} + \omega_r \lambda_{ds}^r \end{cases}$$

$$\begin{cases} \lambda_{ds}^r = L_s i_{ds}^r + \phi_f \\ \lambda_{qs}^r = L_s i_{qs}^r \end{cases}$$

• [Quiz]

Inductance L_s 는 d, q축 flux에 대해 같은가?

flux를 소거하면,

$$\begin{cases} V_{ds}^r = R_s i_{ds}^r + L_s \frac{di_{ds}^r}{dt} - \omega_r L_s i_{qs}^r \\ V_{qs}^r = R_s i_{qs}^r + L_s \frac{di_{qs}^r}{dt} + \omega_r L_s i_{ds}^r + \omega_r \phi_f \end{cases}$$